DEPARTMENT OF MATHEMATICS

MIDTERM TEST #3

MTH 240 – Calculus II

Last Name (Print):_____. First Name:_____. Student Number: _____

Signature:_____.

Date: July 3, 2012, 6:00 pm

Duration: 90 minutes

Section (circle one)

Dr. B. Tasić :	1 2 3 7
Dr. A. Alvarez :	4 5 6 8

Instructions:

1. This is a closed-book test. **Notes, calculators and other aids are not permitted.**
2. Verify that your test has pages 1-6
3. (a) Unless otherwise instructed, **make sure you include all significant steps in your solution, presented in the correct order. Unjustified answers will be given little or no credit. Cross out or erase all rough work not relevant to your solution.** Put a box around your final **answer.**

   (b) Write your solutions in the space provided. If you need more space, use the back of the page. Indicate this fact on the original page, making sure that your solution cannot be confused with any rough work which may be there. Marks (out of 50) are shown in brackets.
4. Do not separate the sheets.
5. Have your student card available on your desk.

For Instructor's use only.

Page	Mark
2	/
3	/
4	/
5	/
6	/
Total	/50

1. [10 marks] Find the radius of convergence and interval of convergence of the series

$$\sum_{n=1}^{\infty} \frac{x^n}{(n^2-2n+3)2^n}$$

Solution: Put $a_n = \dfrac{x^n}{(n^2-2n+3)2^n}$. For $n \geq 1$, $n^2-2n+3 > 0$ and increasing

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left| \frac{x^{n+1}}{(n+1)^2-2(n+1)+3} \cdot \frac{1}{2^{n+1}} \frac{(n^2-2n+3)2^n}{x^n} \right| =$$

$$= \lim_{n\to\infty} \left| \frac{n^2-2n+3}{n^2+2} \cdot \frac{x}{2} \right| = \frac{|x|}{2}$$

3

For $|x| < 2$ $\sum_{n=1}^{\infty} a_n$ is conv

For $|x| > 2$ $\sum_{n=1}^{\infty} a_n$ is div. 1

The radius of convergence is 2 1

For $|x| = 2 \Leftrightarrow x = \pm 2$ we'll test separately.

If $x = 2$ we get $\sum_{n=1}^{\infty} \dfrac{2^n}{(n^2-2n+3)2^n} = \sum_{n=1}^{\infty} \dfrac{1}{n^2-2n+3}$

Putting $b_n = \dfrac{1}{n^2-2n+3}$ and $c_n = \dfrac{1}{n^2}$ we have $\lim_{n\to\infty} \dfrac{b_n}{c_n} = 1 > 0$ and since $\sum_{n=1}^{\infty} c_n = \sum_{n=1}^{\infty} \dfrac{1}{n^2}$ is conv. by the L.C.T $\sum_{n=1}^{\infty} b_n$ is also convergent. 2

If conditions for Tests are not checked −1 mark

If $x = -2$ we get $\sum_{n=1}^{\infty} \dfrac{(-2)^n}{(n^2-2n+3)2^n} = \sum_{n=1}^{\infty} \dfrac{(-1)^n}{n^2-2n+3} = \sum_{n=1}^{\infty} (-1)^n b_n$

So $b_n \to 0$ as $n \to \infty$, $b_n > 0$ for $n = 1, 2, \ldots$ and $b_n \searrow$ for $n \geq 1$. So by the A.S.T $\sum_{n=1}^{\infty} (-1)^n b_n$ is convergent. 2

Finally, the interval of convergence is $[-2, 2]$. 1

Alternatively: To test $x = \pm 2$ one can test $\sum_{n=1}^{\infty} \dfrac{(-1)^n}{n^2-2n+3}$ for abs. conv. It will converge absolutely and that will show that for both $x = \pm 2$ we get a convergent series.

2. [10 marks] Find the Maclaurin series for the function $f(x) = \frac{x}{\sqrt{4+x^2}}$ and its radius of convergence.

(**Hint:** Use the Binomial Series)

$$f(x) = x(4+x^2)^{-\frac{1}{2}} = \frac{x}{2}\left(1+\frac{x^2}{4}\right)^{-\frac{1}{2}} \quad 1$$

$$= \frac{x}{2}\cdot\sum_{n=0}^{\infty}\binom{-\frac{1}{2}}{n}\left(\frac{x^2}{4}\right)^n \quad 2 \qquad \text{for } \left|\frac{x^2}{4}\right| < 1 \iff |x| < 2 \quad 2$$

$$= \frac{x}{2}\sum_{n=0}^{\infty}\frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)\cdots\left(-\frac{1}{2}-n+1\right)}{n!}\frac{x^{2n}}{4^n} \quad 1$$

$$= \frac{x}{2}\sum_{n=0}^{\infty}\frac{-\frac{1}{2}\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\cdots\left(-\frac{2n-1}{2}\right)}{n!}\frac{x^{2n}}{4^n} \quad 1$$

$$= \frac{x}{2}\sum_{n=0}^{\infty}\frac{(-1)^n\,\frac{1\cdot3\cdot5\cdots(2n-1)}{2^n}}{4^n\cdot n!}x^{2n}$$

$$= \frac{x}{2}\sum_{n=0}^{\infty}(-1)^n\frac{1\cdot3\cdot5\cdots(2n-1)}{2^{3n}\cdot n!}x^{2n}$$

$$= \sum_{n=0}^{\infty}(-1)^n\frac{1\cdot3\cdot5\cdots(2n-1)}{2^{3n+1}\cdot n!}x^{2n+1} \qquad \text{for } |x| < 2 \quad 3$$

The radius of convergence is 2

3. [10 marks] a) Approximate $f(x) = \cos 2x$ by a Taylor polynomial $T_4(x)$ with degree 4 at $\frac{\pi}{2}$

$$f(x) = \cos 2x \qquad f\left(\tfrac{\pi}{2}\right) = \cos\pi = -1$$

$$f'(x) = -2\sin 2x \qquad f'\left(\tfrac{\pi}{2}\right) = -2\sin\pi = 0$$

$$f''(x) = -4\cos 2x \qquad f''\left(\tfrac{\pi}{2}\right) = -4\cos\pi = 4$$

$$f'''(x) = 8\sin 2x \qquad f'''\left(\tfrac{\pi}{2}\right) = 8\sin\pi = 0$$

$$f^{IV}(x) = 16\cos 2x \qquad f^{IV}\left(\tfrac{\pi}{2}\right) = 16\cos\pi = -16$$

$$f^{V}(x) = -32\sin 2x$$

0.5 each

2 total

2 total

$$T_4(x) = f\left(\tfrac{\pi}{2}\right) + \frac{f'\left(\frac{\pi}{2}\right)}{1!}\left(x-\tfrac{\pi}{2}\right) + \frac{f''\left(\frac{\pi}{2}\right)}{2!}\left(x-\tfrac{\pi}{2}\right)^2 + \frac{f'''\left(\frac{\pi}{2}\right)}{3!}\left(x-\tfrac{\pi}{2}\right)^3 + \frac{f^{IV}\left(\frac{\pi}{2}\right)}{4!}\left(x-\tfrac{\pi}{2}\right)^4$$

Hence

$$T_4(x) = -1 + \frac{4}{2!}\left(x-\tfrac{\pi}{2}\right)^2 - \frac{16}{4!}\left(x-\tfrac{\pi}{2}\right)^4$$

$$= -1 + 2\left(x-\tfrac{\pi}{2}\right)^2 - \frac{2}{3}\left(x-\tfrac{\pi}{2}\right)^4$$

1

b) Use Taylor's Inequality to estimate the accuracy of the approximation $f(x) \approx T_4(x)$ when $\frac{\pi}{2}-1 \le x \le \frac{\pi}{2}+1$.

1 $\frac{\pi}{2} - 1 \le x \le \frac{\pi}{2} + 1 \iff -1 \le x - \frac{\pi}{2} \le 1 \iff \left|x - \frac{\pi}{2}\right| \le 1$

2 Since $f^{V}(x) = -32\sin 2x$ we have $|f^{V}(x)| \le 32$ for all $x \in \mathbb{R}$

Hence, we can take $M = 32$. Taylor Inequality gives us

2 $|f(x) - T_4(x)| \le \frac{M}{5!}\left|x - \frac{\pi}{2}\right|^5 \le \frac{32}{5!} = \frac{2^5}{1\cdot 2\cdot 3\cdot 4\cdot 5} = \frac{4}{15} \approx 0.26$

4. [10 marks] If $z = f(x, y)$ has continuous second order partial derivatives and $x = r^2 + s^2$ and $y = 2rs$

a) Find $\frac{\partial z}{\partial r}$

$$\frac{\partial z}{\partial r} = \underbrace{\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial r}}_{2} = \underbrace{\frac{\partial z}{\partial x} \cdot 2r + \frac{\partial z}{\partial y} \cdot 2s}_{2}$$

$$\frac{\partial x}{\partial r} = 2r \qquad \frac{\partial y}{\partial r} = 2s$$

b) Find $\frac{\partial^2 z}{\partial r^2}$

$$\frac{\partial^2 z}{\partial r^2} = \frac{\partial}{\partial r}\left(\frac{\partial z}{\partial r}\right) = \frac{\partial}{\partial r}\left(\frac{\partial z}{\partial x} \cdot 2r + \frac{\partial z}{\partial y} \cdot 2s\right)$$

$$= \frac{\partial}{\partial r}\left(\frac{\partial z}{\partial x} \cdot 2r\right) + \frac{\partial}{\partial r}\left(\frac{\partial z}{\partial y} \cdot 2s\right)$$

$$= 2r \frac{\partial}{\partial r}\left(\frac{\partial z}{\partial x}\right) + 2\frac{\partial z}{\partial x} + 2s \frac{\partial}{\partial r}\left(\frac{\partial z}{\partial y}\right) \quad 1$$

if missing -1 mark.

$$= 2\frac{\partial z}{\partial x} + 2r\left(\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)\frac{\partial x}{\partial r} + \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)\frac{\partial y}{\partial r}\right) + 2s\left(\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)\frac{\partial x}{\partial r} + \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)\frac{\partial y}{\partial r}\right) \quad 2$$

$$= 2\frac{\partial z}{\partial x} + 2r\left(\frac{\partial^2 z}{\partial x^2} \cdot 2r + \frac{\partial^2 z}{\partial y \partial x} 2s\right) + 2s\left(\frac{\partial^2 z}{\partial x \partial y} \cdot 2r + \frac{\partial^2 z}{\partial y^2} \cdot 2s\right) \quad 1$$

$$= 2\frac{\partial z}{\partial x} + 4r^2 \frac{\partial^2 z}{\partial x^2} + 4rs \frac{\partial^2 z}{\partial y \partial x} + 4rs \frac{\partial^2 z}{\partial x \partial y} + 4s^2 \frac{\partial^2 z}{\partial y^2} \quad 1$$

$$= 2\frac{\partial z}{\partial x} + 4r^2 \frac{\partial^2 z}{\partial x^2} + 8rs \frac{\partial^2 z}{\partial y \partial x} + 4s^2 \frac{\partial^2 z}{\partial y^2} \quad 1$$

5. [10 marks] Find the local maximum and minimum values and saddle points of the function

$$f(x, y) = x^3 - 12xy + 8y^3.$$

Finding C.Ps

$\frac{\partial f}{\partial x} = 3x^2 - 12y$ (1) ; $\frac{\partial f}{\partial y} = -12x + 24y^2$ (1)

$\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0 \Rightarrow x^2 = 4y$ & $2y^2 = x$

Solving this system

$(2y^2)^2 = 4y \Rightarrow 4y^4 = 4y \Rightarrow 4y(y^3 - 1) = 0 \Rightarrow y = 0$ OR $y = 1$

If $y = 0$ we get $x = 0$

If $y = 1$ we get $x = 2$. So the C.Ps are $(0,0), (2,1)$. (3)

Now we need $\frac{\partial^2 f}{\partial x^2} = 6x$; $\frac{\partial^2 f}{\partial y^2} = 48y$ and $\frac{\partial^2 f}{\partial x \partial y} = -12$ (2)

This gives us $D(x,y) = \frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 = 144(2xy - 1)$ (1)

$D(0,0) = -144 < 0$ and $(0,0)$ is a saddle point (1)

$D(2,1) = 144(2 \cdot 2 \cdot 1 - 1) = 144 \cdot 3 = 432 > 0$ and $\frac{\partial^2 f}{\partial x^2}(2,1) = 6 \cdot 2 = 12 > 0$

So at $(2,1)$ f has a local min $f(2,1) = -8$. (1)